Code Coverage Report

Vehicle Parking System | Team 10 | Software Engineering Mini Project

Statement coverage of the app package, measured with coverage.py while running the full pytest suite

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Build tested	main branch, Team10-VPS/vehicle_parking_system (zip snapshot)
Environment	Python 3.12, Flask 3, Flask-SQLAlchemy, SQLite, pytest 9, coverage 7

1. Summary

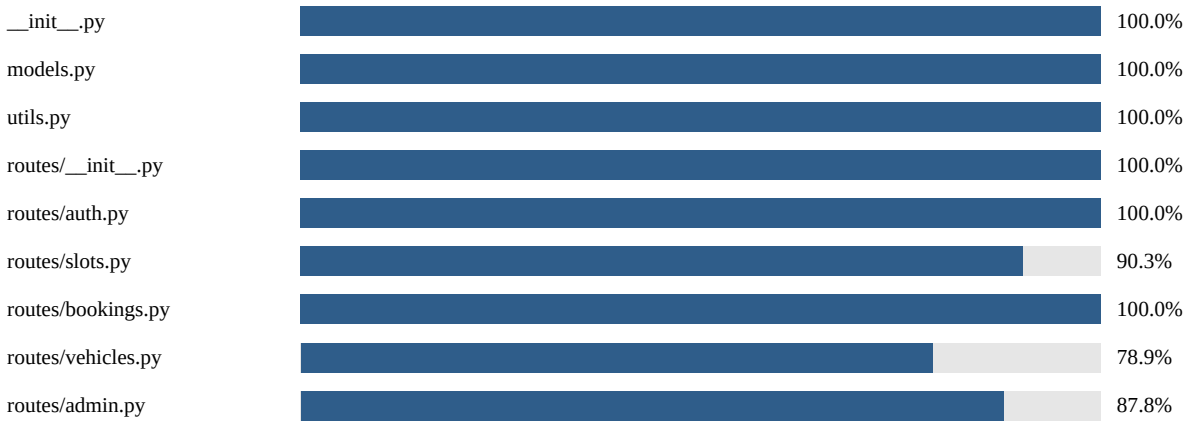
Statements	438
Covered	411
Missed	27
Statement coverage	93.8%

Command used: `coverage run --source=app -m pytest` then `coverage report -m`. All 99 tests passed during the run.

2. Coverage by file

File	Statements	Missed	Coverage	Missing lines
app/__init__.py	51	0	100%	-
app/models.py	40	0	100%	-
app/utils.py	39	0	100%	-
app/routes/__init__.py	0	0	100%	-
app/routes/auth.py	60	0	100%	-
app/routes/slots.py	103	10	90%	32, 33, 35, 36, 67, 68, 79, 100, 102, 118
app/routes/bookings.py	33	0	100%	-
app/routes/vehicles.py	38	8	79%	31, 32, 33, 34, 35, 36, 37, 38
app/routes/admin.py	74	9	88%	63, 71, 72, 73, 77, 106, 107, 108, 109

Statement coverage per file



3. Lines not covered

File	Lines	What the code does
admin.py	63	404 when removing a slot that does not exist

File	Lines	What the code does
admin.py	71-77	Rollback and flash message when slot deletion hits a database error
admin.py	106-109	Rollback and 500 when force check-out fails at commit
vehicles.py	31-38	Duplicate plate and database error handling when registering a vehicle
slots.py	30-36	Booking page for a missing or already occupied slot
slots.py	65-69, 79	Non-numeric slot id handling and invalid vehicle selection message
slots.py	98-102, 118	Plate belonging to another user, type mismatch, slot missing at commit
__init__.py / auth.py	52-58 / 81-87	Secure cookie env setting and config override, logout log line

4. Conclusion

Overall statement coverage is 93.8%. Models, utils and bookings are fully covered. The gaps are mostly error handling paths (database failures, unknown ids) in admin, slots and vehicles, which would need more failure injection tests. No coverage threshold is defined in the Test Plan.